

## CHAPTER 3

# Autism Spectrum Disorders

RAPHAEL BERNIER and GERALDINE DAWSON

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## HISTORICAL CONTEXT

In the early twentieth century Dr. Eugen Bleuler used the term *autism* to describe individuals with schizophrenia who were disconnected from reality ushering in the word into the scientific literature (Bleuler, 1911/1950). Decades later, Dr. Leo Kanner used the term to describe 11 individuals with a pattern of behaviors involving social withdrawal that he felt was distinct from schizophrenia (Kanner, 1943). He referred to this group of patients with challenges in the social and language domains and repetitive behaviors as having early infantile autism. He reported that the patients demonstrated an inability to relate to others as well as social communication challenges including language delays, echolalia, pronominal reversals, and concrete language use. Additionally he described elaborate

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rituals and routines reflecting a significant insistence for sameness (Kanner, 1943).

During the same time period, Dr. Hans Asperger, working in Europe, described a group of boys with similar challenges in social communication (Asperger, 1944/1991). He described a pattern of difficulties in social interaction including impaired use of eye contact, affective expression, and conversational abilities, similar to Kanner's report. However, in contrast to Kanner's patients, the patients Asperger described had well developed language abilities and often spoke pedantically like adults (Asperger, 1944/1991). Although their vocabulary and grammatical skills were intact, these children had significant impairments in conversational skills and had an unusual use of the volume, tone, and rhythm of speech. Finally, Asperger commented on their tendency to become excessively

preoccupied with singular topics of interest, referring to them as little professors because of their tendency to discuss these interests in great detail.

Early theories on the etiology of autism were based on the prevailing atmosphere of psychological understanding of the middle of the twentieth century. At the time psychoanalytic theory was the prominent perspective on the development of mental illness, and although there are many schools of thought within the field, in general, psychoanalytic theory is based on the notion that mental illness arises from unconscious conflicts resulting from early childhood experiences—most often experiences with parental relationships. As such, based on this perspective, if unconscious conflicts resulting from early development are believed to be the root cause of a childhood disorder and if parents, particularly mothers, are the key component of a child's early development, then parents could be considered the causal factor in the development of disorders for which no clear etiology has been identified. Such was the case with autism.

Following his contemporary theorists, Bettelheim (Bettelheim, 1967) proposed that children with autism withdrew from social interaction and turned inward in response to cold, aloof parents—particularly mothers. However, empirical work in the 1970s and 1980s provided evidence counter to this refrigerator mother theory. Parents of children with autism demonstrated personality measures within the normal range and reported similar levels of marital satisfaction and family cohesion relative to parents of children without disabilities (Koegel, Schreibman, O'Neill, & Burke, 1983; McAdoo & DeMyer, 1978). Early scientific leaders arguing against the refrigerator mother theory proposed that the disorder was due to a neurological impairment (Rimland, 1964) and had a strong genetic component (Folstein & Rutter, 1977), and also proposed that involving parents as part of the treatment was more appropriate and beneficial than treating the parents (Schopler & Reichler, 1971).

In the years since Kanner first reported on 11 children who had an "inability to relate themselves in the ordinary way to people and situations from the beginning of life" (Kanner, 1943, p. 242), there have been tremendous gains in our understanding of the core impairments, etiology, and intervention of autism spectrum disorder.

## CORE CHARACTERISTICS

The saying "if you've met one child with autism, you've met one child with autism" speaks to the significant variability in presentation of autism. While each individual with

autism presents with a unique constellation of symptoms and corresponding challenges, there are several characteristics that fall within the domains of social communication and repetitive and restrictive behavior and define the ASD diagnosis.

### Social Attention

From the first days of life typical infants show preferential attention to their social world, attending to faces, voices, and body movements from very early in life. In contrast, young children with autism pay less attention to people and their actions and movements, instead directing their attention to nonsocial objects (Shic, Bradshaw, Klin, Scassellati, & Chawarska, 2011). A primary feature of ASD that emerges by 12 months of age is impaired eye contact (Werner, Dawson, Osterling, & Dinno, 2000; Zwaigenbaum et al., 2005). Toddlers (between the ages of 14 and 42 months of age) with ASD prefer geometric images to images of other children (Pierce, Conant, Hazin, Stoner, & Desmond, 2011), and as opposed to toddlers with either typical development or developmental delays, toddlers with ASD do not show any difficulty disengaging from looking at a face when presented with another stimulus (Chawarska, Volkmar, & Klin, 2010). Two-year-old children with ASD orient to nonsocial contingencies as opposed to biological motion (Klin, Lin, Gorrindo, Ramsay, & Jones, 2009), and preschool-age children with ASD do not show normal preferences for speech sounds (Klin, 1991, 1992; Kuhl, Coffey-Corina, Padden, & Dawson, 2005). Similarly, Ruffman, Garnham, and Rideout (2001) observed intact anticipatory looking during a story task involving nonsocial objects but decreased looking during a story involving human characters in children with ASD compared to children with learning disabilities. This reduced social attention and the observed failure to orient to social stimuli has been termed a *social orienting impairment* and has been noted in preschool-age children with autism and in children as young as 10 months old who go on to develop autism (Dawson, Meltzoff, Osterling, Rinaldi, & Brown, 1998; Werner et al., 2000). In early work in this area, Osterling and Dawson (1994) viewed videotapes of first birthday parties for children who later went on to develop autism and typically developing children and observed that 1-year-olds with autism oriented less often to social stimuli (faces, speech sounds) in their natural environments. In an experimental effort to characterize social attention abilities in ASD, Dawson et al. (Dawson, Meltzoff, Osterling, Rinaldi, et al., 1998) compared rates of orienting to social stimuli (name calling, clapping) and

nonsocial stimuli (rattle, jack-in-the-box) in children with ASD, and receptive language mental age matched children with Down syndrome and typical development. They found that children with ASD more frequently failed to orient to both types of stimuli but this failure was more pronounced for the social stimuli. This body of literature highlighting deficits in social attention suggests that children with ASD may have reduced sensitivity to reward value of social stimuli. As such, these deficits early in development could subsequently result in other, broad downstream social impairments as the child would have reduced opportunities for social learning as a result of the decreased attention and resources devoted to the social world. Indeed, as described later, disruptions to brain circuitry related to reward mechanisms, such as the anterior cingulate cortex (ACC) and orbitofrontal cortex (OFC), have been noted in imaging studies of individuals with ASD indicating differences in neurological functioning of reward related neural networks (Dichter et al., 2010; Haznedar et al., 2000; Schmitz et al., 2008; Scott-Van Zeeland, Dapretto, Ghahremani, Poldrack, & Bookheimer, 2010).

### Joint Attention

Joint attention is defined as the ability to "coordinate attention between interactive social partners with respect to objects or events to share an awareness of the objects or events" (Mundy, Sigman, Ungerer, & Sherman, 1986, pp. 657). By about 9 months of age, typically developing infants begin to look between an object and caregiver and learn to share attention (Walden & Ogan, 1988). This is followed by the incorporation of gesture. By about 12 months of age, infants learn to direct another's attention to an object of interest through gestures such as pointing and also learn to follow another's point (Hannan, 1987). These early emerging joint attention skills are a critical step in the development of spoken language (Bruner, 1975).

Systematic study of joint attention abilities in ASD reveals marked impairments in initiating joint attention, but less so for initiating requests and responding to others' joint attention bids. Children with ASD show less spontaneous sharing or directing of others' attention to an object or action of interest, but less impaired abilities to follow others' gaze and gestures to jointly attend to a point of reference (Mundy, Sigman, & Kasari, 1994; Mundy et al., 1986). Impairments in joint attention, assessed through medical screening visits at 12 and 18 months of age, are present in children later diagnosed with ASD (Baron-Cohen et al., 1996). These joint attention impairments remain consistent over development (Mundy,

Sigman, & Kasari, 1990), are linked with later language abilities (Mundy et al., 1990), and are predictive of subsequent diagnosis and degree of social impairment (Mundy et al., 1994; Sigman et al., 1999; Yoder, Stone, Walden, & Malesa, 2009). Given the prominence of these impairments, a deficit in joint attention is considered a core social impairment in ASD (Charman, 2003; Dawson, Toth, et al., 2004; Loveland & Landry, 1986; Mundy et al., 1986; Sigman et al., 1999; Wetherby & Prutting, 1984). Indeed, joint attention deficits correctly classify 94% of children with ASD in relation to children with intellectual disability (Mundy et al., 1986). Considered in a developmental context, the role of joint attention ability is critical to social and language learning. Indeed, the deficits in this social cognitive ability observed in ASD, as demonstrated in the research literature, predict further downstream impairments later in development.

### Face Perception

At birth infants show a preference for face-like stimuli (Goren, Sarty, & Wu, 1975), and within the first few days of life typical infants are able to recognize their mother's faces (Bushnell, Sai, & Mullin, 1989). Individuals with ASD have consistently shown impairments in face processing abilities. A failure to look at others was the single best discriminator between 12-month-old infants who were subsequently diagnosed with ASD and those going on to develop typically (Osterling & Dawson, 1994), and as such this lack of attention to faces may be one of the most reliable early indicators of ASD risk (Dawson, Webb, & McPartland, 2005). By middle childhood, children with ASD perform worse than mental-age and chronological-age matched peers on tasks of face recognition (Boucher & Lewis, 1992; Boucher, Lewis, & Collis, 1998; Gepner, deGelder, & deSchonen, 1996) and face discrimination (Tantam, Monaghan, Nicholson, & Stirling, 1989). In a controlled study of face processing in ASD, Klin and colleagues (1999) observed impaired face recognition that could not be accounted for by cognitive delays or impairments in spatial memory. In contrast to typical children, children with ASD are better at recognizing isolated facial features and partially obscured faces (Hobson, Ouston, & Lee, 1988; Tantam et al., 1989) show better face memory performance when focusing on the lower half of the face (Langdell, 1978), but show less attention to the core features of the face such as the eyes and nose (Chawarska & Shic, 2009; Klin, Jones, Schultz, Volkmar, & Cohen, 2002; Pelphrey et al., 2002; Trepagnier, Sebrechts, & Peterson, 2002).

This reduced attention to faces may be the downstream effect of reduced social motivation (Chevallier, Kohls, Troiani, Brodtkin, & Schultz, 2012; Dawson, Webb, & McPartland, 2005; Klin et al., 1999). Dawson and colleagues proposed that reduced social motivation results in decreased attention to faces that subsequently results in a failure to develop expertise in the perception of faces. Without this experience expectant environment, the brain systems needed to process faces in a typical pattern do not develop (Dawson, Webb, & McPartland, 2005).

Indeed, several studies indicate atypical neural functioning in face perception in ASD. Young children with ASD show an atypical faster neural response using neurophysiological assessment to objects than to faces as opposed to the opposite pattern observed in typical children (Webb, Dawson, Bernier, & Panagiotides, 2006). Preschool-age children with ASD show neural responses to familiar and unfamiliar faces similar to typically developing toddlers, suggesting delayed development in face processing circuitry (Webb et al., 2011). Slower neural responses to faces have been observed in adolescents and adults with ASD (McPartland, Dawson, Webb, Panagiotides, & Carver, 2004; O'Connor, Hamm, & Kirk, 2007), parents of children with ASD (Dawson, Webb, Wijsman, et al., 2005), and 10-month-old infants at risk for ASD (McCleery, Akshoomoff, Dobkins, & Carver, 2009), suggesting that the neural systems underlying face processing may be less efficient and lack specificity to faces. However, not all results have been consistent (for a review, see Dawson, Webb, & McPartland, 2005).

### Emotion Perception

By 6 months of age typical infants differentially respond to the emotional displays of those in their environment (Trevvarthen, 1979). That is, they smile more and direct more attention to happy faces relative to neutral or sad expressions (Rochat & Striano, 1999; Tronick, Als, Adamson, Wise, & Brazelton, 1978). By the time infants are between 9 and 12 months of age they have learned to utilize others' facial expressions to guide their own behavior when in unfamiliar situations or presented with novel stimuli (Feinman, 1982; Moore & Corkum, 1994). The ability to empathically respond to another's distress also begins to emerge within the first years of life, and is reflected by demonstrations of concern, sharing, helping, and comforting (Rheingold, Hay, & West, 1976; Zahnwaxler & Radkeyarrow, 1990).

In his initial report in 1943, Kanner noted that individuals with ASD are impaired in the recognition of affective

cues. A number of empirical studies have demonstrated this observation through carefully controlled experimentation. Studies have shown that children with ASD fail to attend to others' affective displays normally (Bacon, Fein, Morris, Waterhouse, & Allen, 1998; Blair, 1999; Charman et al., 1998; Dawson, Meltzoff, Osterling, & Rinaldi, 1998; Hall, Szechtman, & Nahmias, 2003; Sigman, Kasari, Kwon, & Yirmiya, 1992) and show deficits in recognizing emotional expressions (Bormannkischkel, Vilsmeier, & Baude, 1995; Braverman, Fein, Lucci, & Waterhouse, 1989; Hobson, 1986a, 1986b; Macdonald et al., 1989). Pioneering work in which a familiar examiner pretended to hurt himself or herself by hitting one finger with a plastic hammer and then displayed facial and vocal expressions of distress without using words demonstrated differential response to distress. Sigman and colleagues (1992) observed the behavioral responses of 3- to 4-year-old children with ASD, intellectual disability, and typical development and found that children with ASD often failed to notice or ignored the examiner's affective displays, while the comparison groups were very attentive to the examiner's affective display regardless of the type of emotion displayed. However, employing a similar design, Dawson and colleagues (1998) found that although still attending less than controls, the children with ASD were able to distinguish between emotions; they were more likely to observe and attend to distress relative to a neutral expression. This reduced attention to emotional expressions has also been noted in children with ASD as young as 2 years old (Charman et al., 1998).

The behaviorally observed differences in emotion perception are reflected in observed neurophysiological differences as well. Dawson and colleagues (Dawson, Webb, Carver, Panagiotides, & McPartland, 2004) assessed neural responses to images of a fear face and a neutral face in 3–4-year-old children with ASD as well as typically developing, age matched peers, and chronological and mental aged matched children with developmental delay using event related potentials. During passive viewing of facial expressions the toddlers with ASD failed to demonstrate the increased amplitude to the fear face that was observed in the typical children. Importantly, delays in neural processing of the fear face were associated with reduced social attention assessed through experiments conducted on a different day. Specifically, greater delays in processing the emotional expressions were associated with decreased attention to an experimenter expressing distress, fewer joint attention bids, and reduced social orienting. Importantly, these affective processing delays were not correlated with task performance involving attention to

nonsocial stimuli, such as orienting to nonsocial sounds. This study suggests that children with ASD show deficits in basic aspects of emotion perception by at least 3 years of age and that these challenges are associated with poor social abilities more broadly.

### Imitation

Imitation is an early emerging social ability. Newborns as young as 42 minutes old can imitate facial expressions (Meltzoff & Moore, 1977, 1979, 1983) and from there, these early skills develop rapidly over the first years of life with the imitation of actions on objects and imitation of simple gestures, such as waving, emerging around one year of age. Imitation skills progress from simple and familiar actions to increasingly complex, novel, and sequenced behaviors (Meltzoff & Moore, 1994; Piaget, 1962) with the application of imitative actions in novel contexts, emerging between the first and second year, although some researchers suggest the skill emerges earlier (see Bauer, 1996, for review).

It has been proposed that imitation serves as a foundation for social reciprocity and connectedness with others as well as the child's ability to differentiate self from others (Eckerman, Davis, & Didow, 1989; Meltzoff & Gopnick, 1993; Nadel, Guerini, Peze, & Rivet, 1999; Trevarthen, Kokkinaki, & Fiamenghi, 1999; Uzgiris, 1981), developing symbolic play (Piaget, 1962), peer relationships (Trevarthen et al., 1999), language (Avikainen, Wohlschlager, Liuhanen, Hanninen, & Hari, 2003; Charman et al., 2003), and emotional sharing (Hatfield, Cacioppo, & Rapson, 1994). It has also been proposed to play a role in understanding about other's intentions and goals (Kugiumutzakis, 1999; Uzgiris, 1999). In showing that 18-month-old children can imitate others' intended but not completed actions, Meltzoff (1995) proposed that imitation allows a child to understand others' internal states and through the process of learning that the child can produce the same action sequence as what he or she has just observed is a precursor for the development of a theory of mind (Meltzoff & Gopnick, 1993; Rogers & Pennington, 1991).

DeMeyer and colleagues (1972) first reported imitation deficits in ASD after studying nine children with ASD and 12 age-matched children with intellectual disability. The individuals with ASD performed significantly poorer on a large battery of tasks assessing imitation of body movements and actions on objects. The children did not differ in their ability to perform non-imitated actions on objects, suggesting it was imitation and not motor deficits accounting for the observed differences.

Despite the varied testing methodologies, sample characteristics, and comparison groups employed in the studies conducted in the years since DeMeyer and colleagues' report, imitation impairments in ASD have consistently been noted (Williams, Whiten, Suddendorf, & Perrett, 2001). Impairments in the ability to imitate the others' movements, including simple body movements and actions with objects have been observed in young children with ASD (Curcio, 1978; Dawson & Adams, 1984; Dawson, Meltzoff, Osterling, & Rinaldi, 1998; DeMeyer et al., 1972; Sigman & Ungerer, 1984; Stone, Ousley, & Littleford, 1997). Deficits in both immediate and deferred imitation have been noted (Dawson, Meltzoff, Osterling, & Rinaldi, 1998) as well in both *procedural* imitation (i.e., the imitation of actions on novel objects) and *gestural* imitation (i.e., the imitation of meaningless gestures), although gestural impairments were much greater for the autism group (Roeyers, Van Oost, & Bothuyne, 1998). Hobson and Lee found no differences between teens with ASD and age matched peers with intellectual disability in imitation of goal directed actions. However, fewer adolescents with ASD accurately imitated the style with which the experimenter performed the observed action than the control group. They concluded that imitation impairments in ASD reflect a deficit in "the coordination of the representation between the self and other" (Hobson & Lee, 1999). Using a manualized coding system, Bernier and colleagues (Bernier, Dawson, Webb, & Murias, 2007) enlisted trained raters naïve to subject diagnosis to code imitation ability across single step and sequenced behavioral movements in adults with ASD and normal cognitive ability. The adults performed worse on imitative tasks and made more errors in completing the imitative tasks relative to typical peers suggesting imitative deficits are not simply the result of delays in development.

Although there is a general consensus that imitation ability is impaired in ASD, the underlying etiology of particular deficit is still under debate. Theories have attributed imitation impairments to motor praxis impairments and to planning, sequencing, and executing intentional motor movements, although individuals with motor deficits perform better on imitative tasks than ASD (Green et al., 2002; Rogers, 1998). Others have proposed that difficulties in understanding the intersubjective experience of others leads to impairments in matching the movements of self and others (Hobson & Lee, 1999; Meltzoff & Gopnick, 1993). Another theory suggests inadequate procedural learning and inefficient cerebellar-basal ganglia-parietal circuitry contribute to the imitation impairments (Mahone et al., 2006; Mostofsky, Goldberg, Landa, & Denckla,

2000) while neurophysiological and brain imaging studies suggest that a dysfunctional mirror neuron system may play a critical role in imitation deficits (Bernier, Aaronson, & McPartland, 2013; Bernier et al., 2007; Dapretto et al., 2006; Iacoboni & Mazziotta, 2007; Martineau, Cochin, Magne, & Barthelemy, 2008; Oberman et al., 2005; Oberman, Ramachandran, & Pineda, 2008; Rizzolatti & Craighero, 2004; Williams et al., 2006). Regardless of the etiology of the imitation impairments in ASD, given the prominent role of imitative ability to subsequent social cognitive abilities, it is clear that early disruption to imitation can result in deviations from the normal social cognitive developmental tract and result in further downstream developmental challenges for children with ASD.

### Symbolic Play

Symbolic representation through pretend play is an important precursor to language development that gradually emerges in the second year of life (Riguet, Taylor, Benaroya, & Klein, 1981; Ungerer & Sigman, 1984). By approximately 20 months of age most children can engage in play in which they attribute animate characteristics to inanimate objects (e.g., pretending that a doll can talk) and can use one object as if it were another (e.g., pretending that a block is a car).

Early reports indicate impairments in symbolic play in ASD. Wulff (1985) stated "the autistic child's play is striking in its lack of fantasy or all other aspects of symbolic play" (pp. 146). Indeed, a key indicator for children who went on to develop ASD noted through a prospective medical screening study was the absence of pretend play at 18 months of age in (Baron-Cohen et al., 1996). Importantly, as their language develops, individuals with ASD show increases in their symbolic play (Amato, Barrow, & Domingo, 1999), and targeted interventions focusing on improving symbolic play yield promising results (Kasari, Freeman, & Paparella, 2006). In scaffolded situations children with ASD can engage in symbolic play (McDonough, Stahmer, Schreibman, & Thompson, 1997), but spontaneous symbolic play is limited, mechanical and repetitive (Wing & Gould, 1978), more akin to learned routines than elaborate, complex, pretend stories (Williams, Reddy, & Costall, 2001). It is unclear if the deficits in symbolic play are the result of a metarepresentational impairment in attending, understanding, and imitating others' actions on objects, that is, a function of joint attention impairments or a function of a generalized cognitive deficit in symbolic thinking, along the lines of an executive function deficit (see (T. Charman, 1997), for a review). Regardless of the

cause of the symbolic play deficits, the child with ASD who lacks the opportunity for language and representational learning practiced through play will miss out on the needed opportunities for learning in this arena, resulting in challenges that will only compound with development.

### Language and Communication

Given the deficits in the above mentioned early-emerging social communication abilities that are precursors to language development (e.g., joint attention), it is not surprising that many children with ASD have significantly language delays and atypical language development. In fact, language delay is often cited as the initial concern for many parents whose children are later diagnosed with ASD (Chawarska et al., 2007; De Giacomo & Fombonne, 1998; Siklos & Kerns, 2007).

Although it has been historically reported that approximately 50% of individuals with ASD remain nonverbal throughout their lives (Rutter, 1978), this estimate has decreased considerably with advances in earlier diagnosis and intervention. Anderson and colleagues (2007) assessed 9-year-old children with ASD and found that approximately 25% of the children were effectively nonverbal, and another 25% were fluent speakers using complex sentence structure. Findings from longitudinal studies suggest that variability in language ability increases over time, possibly due to responses to intervention (Charman et al., 2005).

Pioneering work in early social communication found that those children with ASD who developed gestural, nonverbal joint attention skills were more likely to develop language (Mundy et al., 1990; Sigman et al., 1999) and a prospective study following high-risk infants who later went on to develop ASD fewer joint attention and gestural behaviors indicating that these early social communicative abilities may be key behavioral indicators of ASD and later language ability (Landa, Holman, & Garrett-Mayer, 2007).

The presence of atypical language is often noted in individuals ASD and limited language. Many individuals with ASD have atypical language characterized by immediate or delayed echolalia (e.g., verbatim repetition of previously heard words or phrases; Rydell & Mirenda, 1994), atypical prosody (e.g., abnormal rhythm, stress, intonation, and loudness; Peppe, McCann, Gibbon, O'Hare, & Rutherford, 2007), neologisms or idiosyncratic language (e.g. the use of made of words; Volden & Lord, 1991), and pronominal reversal (e.g., use of "you" instead of "I" when referring to the self; Cantwell, Baker, Rutter, & Mawhood, 1989; Kanner, 1943).

Despite the presence of language delays and atypical use of language, the most pronounced language impairments in ASD fall in the pragmatic, or social, aspects of language use (see Tager-Flusberg, 1999, 2001, for reviews). Noted characteristics of conversational exchanges with individuals with ASD include the use of irrelevant detail in conversations (e.g., providing dates and ages when discussing a particular event or person), perseveration of particular topics, pedantic speaking on a particular topic of conversation, inappropriate shifts to a new topic without context or consideration of the listener, and a lack of responsiveness to conversational topic initiations introduced by another person (Eales, 1993; Paul, Orlovski, Marcinko, & Volkmar, 2009; Tager-Flusberg, 1999, 2001). These challenges, along with reduced responses to questions and fewer spontaneous offerings of information disrupt the sense of reciprocity in conversational exchanges (Capps, Kehres, & Sigman, 1998). These difficulties with the social aspect of language are also observed among highly verbal adults with ASD (Lord & Paul, 1997). It has been proposed that these pragmatic deficits are the result of impairments in understanding another's intention during conversations (Eales, 1993) or because individuals with ASD have difficulty taking the listener's perspective into account (Tager-Flusberg, 1993, 1996). This difficulty is reflected in atypical language use such as abnormal pronoun use (e.g., saying "You want a drink" when meaning "I want a drink") and inaccurate application of questions and statements (e.g., saying "Want milk?" instead of stating "I want milk").

Language comprehension difficulties have been noted in ASD as well. Many individuals have difficulty understanding non-literal language such as irony, sarcasm, and metaphor (Happé, 1995; MacKay & Shaw, 2004; Martin & McDonald, 2004). Studies also suggest that individuals with ASD also struggle to understand vocal utterances based on stress patterns (Peppe et al., 2007) or identify affective content in a voice (Korpilahti et al., 2007; Rutherford, Baron-Cohen, & Wheelwright, 2002). Rapin (1996) noted that many individuals with ASD seem to have greater challenges in understanding other people's language than they do in learning the specific language structures necessary to produce language (Rapin, 1996). For example, in a study of children with ASD compared to children with specific language impairment, the children with ASD tended to rely on a syntactically based word order strategy rather than relying on the semantic content to determine the meaning of a sentence (Rhea Paul, Fischer, & Cohen, 1988).

It is becoming clear that the language delays and atypical language observed in ASD often result from and interact with other deficits such as comorbid language processing deficits (Bishop & Norbury, 2002; Rapin & Dunn, 2003; Tager-Flusberg, 2004); difficulties in theory of mind (Paul et al., 2009; Tager-Flusberg, 2000); and repetitive behavior being reflected in stereotypical speech patterns (Bodfish, Symons, Parker, & Lewis, 2000; Gabriels, Cuccaro, Hill, Ivers, & Goldson, 2005).

### Restricted and Repetitive Interests and Behaviors

Restricted and repetitive interests and behaviors (RRBs) are considered a core diagnostic feature of ASD (Lord & Luyster, 2006; Lord et al., 2006) and were identified as part of Kanner's initial report on ASD. RRBs include repetitive, stereotypic motor mannerisms such as hand flapping, finger flicking, or complicated whole body movements such as posturing or spinning and repetitive use of objects, such as lining things, spinning the wheels of a toy car, playing with strings, repeatedly dropping something from the same height. RRBs also include interests that are unusual in their focus, such as an interest in street signs or toilets, or in their intensity, such as a focused preoccupation on a particular topic typically involving the memorization of facts (e.g., train timetables). An insistence on sameness, reflected in difficulties with transitions or changes to routines or by the presence of ritualized sequences of actions or the precise arrangement of space also falls under the category of RRBs. Finally, sensory interests, such as interests in the sight, sound, smell of things or people (e.g., interest in hair) and aversions (e.g., aversions to specific sounds or particular textures, sound sensitivities necessitating removal from stimuli or situation) are also components of this category.

The observed sensory related challenges can be subdivided to reflect sensory over-responsivity (e.g., negative responses to specific sensory stimuli such as lights or sounds), sensory underresponsivity (e.g., reduce responsiveness to various sensory stimuli), and sensory seeking (e.g., actions performed to increase sensation) (Dunn, 1997; Miller, Anzalone, Lane, Cermak, & Osten, 2007). All three of these sensory processing profiles have been observed in individuals with ASD (Baranek, David, Poe, Stone, & Watson, 2006). Rates of sensory overresponsivity in ASD range from 56 to 70% (Baranek et al., 2006; Ben-Sasson et al., 2007) with common manifestations including distractibility because of the noise emitted by fluorescent lights or a distant siren. However, the same

children with overresponsivity can at times show underresponsivity such as in the form of insensitivity to pain. It has been suggested that impaired ability to modulate sensory information underlies these challenges observed in ASD (Ornitz, 1989).

It has been proposed that RRBs in ASD fall in two distinct categories: lower level behaviors including repetitive motor movements and higher level behaviors that are characterized by insistence on sameness and circumscribed interests (Turner, 1999). The lower level behaviors appear to occur more often in younger children with ASD or children with comorbid intellectual impairments (Wing, 1988; Wing & Gould, 1979), while the higher level behaviors are observed in children without significant intellectual impairments. Both forms of RRBs appear by 2 to 3 years of age (Bishop, Richler, & Lord, 2006; Honey, McConachie, Randle, Shearer, & Le Couteur, 2008; Militeri, Bravaccio, Falco, Fico, & Palermo, 2002; Mooney, Gray, & Tonge, 2006) and longitudinal studies suggest that they persist into adolescence and adulthood (Piven, Harper, Palmer, & Arndt, 1996; Rumsey, Rapoport, & Sceery, 1985).

No specific type of repetitive behavior is specific to ASD as RRBs are observed in typically developing infants and young children (e.g., Evans et al., 1997; Thelen, 1979), and in other neurodevelopmental and psychiatric disorders (Frith & Done, 1990). For example preferences for routines and distress from changes to expectations are observed in typically developing young children (Evans et al., 1997) and in individuals with psychiatric disorders, such as obsessive-compulsive disorder, and repetitive mannerisms are often seen in individuals with intellectual disability. However, relative to typical development in which these behaviors happen fleetingly and to other disorders, in ASD these behaviors occur more frequently (Matson, Dempsey, & Fodstad, 2009). Indeed, the observed pattern of multiple types and severity of RRBs can aid in the discrimination of ASD from other disorders (Bartak & Rutter, 1976; Bodfish et al., 2000; Charman & Swettenham, 2001).

More research is needed to elucidate the relationship between RRBs and the social communication impairments characterizing ASD. Historically, studies examining early emerging ASD characteristics suggested that social communication impairments were the first indicators to emerge in ASD (Baron-Cohen et al., 1996; Cox et al., 1999; Osterling & Dawson, 1994; Robins, Fein, Barton, & Green, 2001); however, recent prospective studies of at risk infant siblings of children with ASD suggest that RRBs, including spinning and spending more time attending to objects observed at 12 months of age are predictive of subsequent

ASD symptoms and diagnosis (Ozonoff, Heung, Byrd, Hansen, & Hertz-Picciotto, 2008; Zwaigenbaum et al., 2005). While some argue for an underlying continuous factor of ASD symptoms rather than distinct domains of impairment (Constantino et al., 2004), others cite twin studies showing that these ASD domains are only modestly correlated and conclude they are likely distinct (Happé, Ronald, & Plomin, 2006). The presence of circumscribed interests may provide insight. These intense, perseverative interests usually involve memorization of facts about a specific topic (e.g., cameras, trains) that are nonsocial in nature (South, Ozonoff, & McMahon, 2005) and appear to be common to ASD (one study found that approximately 88% of children with ASD had circumscribed interests) regardless of intellectual functioning or symptom severity (Lam, Bodfish, & Piven, 2008). The nonsocial focus of these interests could diminish motivation and opportunities for social experiences, such that decreased reward salience for social stimuli and increased valence for non-social stimuli could result in the sequelae of impairments observed across both domains in ASD.

## DIAGNOSIS AND ASSESSMENT OF ASD

There is currently no biological marker or medical test for making a diagnosis of ASD. The diagnosis is made through careful evaluation of developmental history and caregiver report of symptomatology and observation of the presence (or absence) of behaviors across the two domains of ASD: social communication and repetitive and restricted behavior. The two widely used systems for diagnosing ASD include the *Diagnostic and Statistical Manual of Mental Disorders*, 5th ed. (*DSM-5*; American Psychiatric Association, 2013) and the *International Classification of Diseases*, 10th ed. (*ICD-10*; World Health Organization, 1992). The specific criteria are described in Table 3.1.

### Changes in the Diagnosis

The *DSM-5* was revised in 2013 following nearly two decades of research into ASD (American Psychiatric Association, 2013). With the revised *DSM*, the previous diagnostic umbrella term *pervasive developmental disorders* is no longer used, and the three diagnostic subtypes of autistic disorder, Asperger's disorder, and pervasive developmental disorder—not otherwise specified are collapsed into one term: autism spectrum disorder. Although controversial, the *DSM* committee's decision to collapse all three diagnostic subtypes under one diagnosis was

**TABLE 3.1** *DSM-5 Diagnostic Criteria for Autism Spectrum Disorder*

- (A) Clinically significant, persistent deficits in social communication and interactions across multiple settings, currently or by history, as manifested by all of the following:
1. Deficits in social-emotional reciprocity
  2. Marked deficits in nonverbal communication used for social interaction
  3. Failure to develop, maintain, and understand relationships
- (B) Restricted, repetitive patterns of behavior, interests, and activities, currently or by history, as manifested by at least two of the following:
1. Stereotyped or repetitive motor movements, actions on objects, or speech
  2. Insistence on sameness and excessive adherence to routines or ritualized patterns of behavior
  3. Highly restricted, fixated interests
  4. Hyper- or hyporeactivity to sensory input or unusual sensory interests
- (C) Symptoms must be present in early childhood (but may not become fully manifest until social demands exceed limited capacities)
- (D) Symptoms must cause clinically significant impairment in social, occupational, or other areas of functioning

based on findings that no specific etiology was tied to the subtypes, no treatment was specific to each subtype, and even experienced clinicians were unable to agree on which children should be diagnosed with each subtype (King, Navot, Bernier, & Webb, 2014; Lord et al., 2011). Other changes included the collapse of three domains of impairment (social, communication, and restricted/repetitive behaviors) into two domains (social communication and restricted/repetitive behaviors). The collapsing of the domains was to reflect that all communication is social in nature and that certain symptoms under the previous system could be counted in both the social domain and the communication domain giving additional weight to a particular symptom (King et al., 2014). Finally, qualifiers, a severity rating system, the inclusion of sensory sensitivities to the RRB domain, and the removal of the requirement that deficits be observed by 3 years of age were all changes made during this revision process.

### Domains of Impairment, Qualifiers, and Severity

As shown in Table 3.1, to meet diagnostic criteria for ASD, a child must have deficits in three areas in the social communication domain, at least two of the four possible symptoms in the RRB domain. These impairments may not become evident until the social demands exceed the capacities of the individual and to reflect that there has been a shift from the previous requirement of symptoms present prior to three

years of age to an allowance for diagnosis later in life and even adulthood.

With the adoption of the *DSM-5* criteria, qualifiers have been appended to the diagnostic description for an individual. Due to the commonality of the behaviors seen across the *DSM-IV* ASD subtypes, the *DSM-5* committee concluded that the constellation of impairments observed in ASD are best represented as a single yet multidimensional disorder that manifests it uniquely for each individual due to differing cognitive and language abilities, and comorbid disorders (Rutter, 2011). As such, qualifiers to the ASD diagnosis include the presence of co-occurring intellectual or language impairment and association with a known medical or genetic condition or environmental factor. Severity levels have also been incorporated into the *DSM-5* diagnostic process with three levels of support defined for each ASD domain. The levels range from level 1, requiring support, which reflects a severity level such that if a child were without supports, the child's deficits in social communication would cause noticeable impairments spanning level 2, requiring substantial support, and progressing to level 3, requiring very substantial support, which for the social communication domain is defined as deficits in verbal and nonverbal social communication skills that cause severe impairments in functioning (American Psychiatric Association, 2013).

## RELATED SYMPTOMS AND COMORBID DISORDERS

### Intellectual Disability

Current estimates of the rate of intellectual disability (i.e., IQ below 70) in individuals with ASD are at approximately 40% with higher rates of comorbidity observed in girls with ASD (Baird et al., 2000; Centers for Disease Control and Prevention, 2009, 2012; Chakrabarti & Fombonne, 2001). These estimates differ from earlier estimates published between 1966 and 1999, reviewed by Fombonne (1999), which ranged from 44 to 100% with rates of individuals without intellectual disability estimated at 25.4%, mild to moderate impairment at 23.2%, and severe to profound mental impairment estimated at 55.5%. The decline in the rate of comorbid intellectual disability among individuals with ASD is likely a function of increased diagnostic rates of individuals with intact cognitive abilities along with the effectiveness of early intervention (Chakrabarti & Fombonne, 2005; Fombonne, 2003a, 2003b; Matson & Shoemaker, 2009; Newschaffer, Falb, & Gurney, 2005a).

The diagnostic differential between intellectual disability and ASD lies most clearly in the display of specific impairments in joint attention, motor imitation, symbolic play, theory of mind, and expressive versus receptive language skills (Maljaars, Noens, Scholte, & van Berckelaer-Onnes, 2012; Poon, Watson, Baranek, & Poe, 2012; Wong & Kasari, 2012) observed in ASD. Although some investigators report a specific pattern of intellectual abilities such that individuals with ASD perform better on nonverbal visual-spatial tasks relative to verbal tasks (Fombonne, 2003b; Happe, 1994; Lockyer & Rutter, 1970; Matson & Shoemaker, 2009), others have failed to observe this (Mayes & Calhoun, 2003; Minshew, Turner, & Goldstein, 2005; Siegel, Minshew, & Goldstein, 1996; Szatmari, Tuff, Finlayson, & Bartolucci, 1990), or note that this cognitive profile is not consistent across the age span (Ankenman, Elgin, Sullivan, Vincent, & Bernier, 2014).

### Communication Disorders

Given the previously reported deficits in language and communication, communication disorders are often diagnosed at some point in the diagnostic journey for an individual with ASD. Many of the language impairments observed in ASD are reflected in the diagnosis of language disorder, characterized by deficits in comprehension or production of language, although the early emerging social deficits and presence of stereotypic interests or behaviors can readily differentiate language disorder from ASD (American Psychiatric Association, 2013). Speech sound disorders, defined by persistent deficits in speech sound production can also be diagnosed in many children with ASD and may be linked to common underlying etiologies. For example, as described already, many individuals with 16p11.2 deletion have speech sound disorders, but this also one of the most common risk factors for ASD, such that many children with 16p11.2 deletion have both ASD and speech sound disorders (Hanson et al., 2014).

The *DSM-5* also includes a new diagnostic classification related to communication impairment: social communication disorder (SCD). The *DSM-5* defines SCD as persistent difficulty with verbal and nonverbal communication, such as difficulties with spoken and written language and conversational impairments, that is not better accounted for by cognitive impairments. These social communication impairments overlap with many ASD symptoms rendering the differential diagnosis difficult; however, the SCD diagnosis specifies the absence of repetitive or stereotyped behaviors. The lack of repetitive or restricted behaviors

differentiates ASD and SCD. As such, ASD must be ruled out before making a diagnosis of SCD.

### Attention-Deficit/Hyperactivity Disorder

Prior to the *DSM-5*, attention-deficit/hyperactivity disorder (ADHD) and ASD were considered exclusive of one another and a diagnosis of ASD precluded a diagnosis of ADHD (American Psychiatric Association, 2000). However, some individuals do show symptoms of ADHD, including inattention, hyperactivity, and impulsivity, and the *DSM-5* now allows for the presence of both conditions in one individual (Mayes, Calhoun, Mayes, & Molitoris, 2012). Over the past 10 years, study of the overlap between the two disorders has expanded considerably with a focus on common etiology, similar neurological origins, the evidence for comorbid occurrences, and the implications of a dual diagnosis. Common genetic events have been identified between both disorders (Williams et al., 2010) and decreased prefrontal cortex activation has been reported in both ASD and ADHD (Happé, Booth, Charlton, & Hughes, 2006), providing putative biological bases for the overlap in symptom presentation. Indeed, rates of comorbidity across ASD and ADHD range from 33 to 78% (Gargaro, Rinehart, Bradshaw, Tonge, & Sheppard, 2011; Goldstein & Schwebach, 2004; Sinzig, Walter, & Doepfner, 2009). Interestingly, for those individuals dually diagnosed with ASD and ADHD, oppositional behavior, but not anxiety or ADHD symptom severity is more often reported (Grzadzinski et al., 2011).

### Anxiety and Mood Disorders

Co-occurring anxiety and mood disorders are commonly reported in individuals with ASD (Strang et al., 2012; Whitehouse, Durkin, Jaquet, & Ziatas, 2009; Whitehouse, Watt, Line, & Bishop, 2009), however, it is sometimes difficult to identify these symptoms, especially in individuals with ASD who have poor ability to report their internal states. For example, social phobia and social withdrawal may be manifestations of anxiety and depression, respectively, or the result of social impairments related to ASD. Given the clinician's reliance on self-report of symptoms in the assessment of anxiety and mood disturbances and the observation that individuals with ASD often have difficulty identifying and describing their own emotions and feelings (Gillott, Furniss, & Walter, 2001), the ability to detect these symptoms is sometimes challenging. The clinician often must rely on parent or supplemental reports

(Stewart, Barnard, Pearson, Hasan, & O'Brien, 2006), which are known to be less reliable than self-report for both disorders. Further, it is difficult at times to determine if these symptoms are actually distinct diagnoses or rather a characteristic of the disorder (e.g., obsessive compulsive symptoms). Importantly, both anxiety and depression are both responsive to pharmacological and behavioral intervention in individuals with ASD.

Depression in ASD often manifests as an increase in agitation, social withdrawal, and compulsions along with changes in sleep and appetite. Rate estimates for comorbid depression and ASD range from 1 to 58% (Lainhart, 1999; Mayes, Calhoun, Murray, Ahuja, & Smith, 2011). Studies suggest that symptoms of depression are often observed during adolescents in individuals with ASD who have greater insight into their social communication challenges, behavioral differences show an increase in wanting to develop friendships (Kim, Szatmari, Bryson, Streiner, & Wilson, 2000; Mayes et al., 2011).

Research in depression and ASD suggests that even though an individual with ASD may not meet full diagnostic criteria for a comorbid depressive disorder, the manifestation of depressive symptoms can still significantly impact his or her quality of life (Leyfer et al., 2006a; Simonoff et al., 2008). Interestingly, it has been noted that the rate of suicidal ideation is less likely in individuals with ASD and depression such that suicidal ideation is 2.6 times more likely to occur in children with depression alone than in children with both disorders (Mayes et al., 2012; Stewart et al., 2006). A systematic review of several large-scale studies, however, indicated that individuals with ASD comprised 7.3–15% of suicidal populations. This is a substantial subgroup. Several risk factors for suicide among individuals were also reported including peer victimization, behavioral problems, Black or Hispanic ethnicity, lower socioeconomic status, and lower level of education (Segers & Rawana, 2014).

Difficulties with the assessment and characterization of anxiety in ASD have made establishing the prevalence of co-occurrence challenging and as a result study estimates have varied considerably (Leyfer et al., 2006a; Mayes et al., 2012; Simonoff et al., 2008). Studies report ranges from 7 to 84% with the most commonly co-diagnosed disorders being generalized anxiety disorder, agoraphobia, and simple phobia (Gillott et al., 2001; Lainhart, 1999; Leyfer et al., 2006b). Challenges in the symptom overlap make disentangling the disorders difficult. For example, symptoms of obsessive-compulsive disorder (OCD) can resemble the ritualized and repetitive behavior

observed in ASD. Although OCD can be differentiated from ASD in that individuals with OCD are more often fixated on a single compulsion than individuals with ASD (Russell, Mataix-Cols, Anson, & Murphy, 2005; Russell & Sofronoff, 2005; Wakabayashi, Baron-Cohen, & Ashwin, 2012), this requires careful clinical investigation. And, similar to depression, certain subgroups of ASD may be more prone to anxiety disorders such that adolescents with intact cognitive abilities may be more likely to have comorbid anxiety than others (Kim et al., 2000; Mattila et al., 2010; Mayes et al., 2011).

### Self-Injurious Behaviors

Self-injurious behaviors can include head banging, head slapping, hair pulling, and finger or hand biting and have been observed in individuals with ASD. However, the presence of these behaviors may be more linked to cognitive and language deficits in ASD, rather than ASD itself (Dawson, Matson, & Cherry, 1998). It has been suggested that self-injurious behaviors can be the expression of frustration in individuals with limited communication ability to express feelings or needs (Donnellan, Mirenda, Mesaros, & Fassbender, 1984; Lainhart, 1999). Regardless of the origin, self-injurious behaviors are associated with increased perceived stress in families (Bishop, Richler, Cain, & Lord, 2007); however, behavioral treatments can prove effective in reducing these behaviors.

### Seizures

Epilepsy is diagnosed in 6–50% of individuals with ASD depending on the study source with population-based studies reporting lower rates of 5% and epilepsy specific clinics citing rates of 15–30% (Clarke et al., 2005; Matsuo, Maeda, Sasaki, Ishii, & Hamasaki, 2010). Seizure onset most often occurs either early in childhood or in adolescence (Parmegiani et al., 2010) with early onset seizures associated with poorer outcomes (Saemundsen, Ludvigsson, & Rafnsson, 2008). Given the increased likelihood for seizures in ASD, it is recommended that individuals with ASD suspected of having seizures be promptly evaluated (Filipek et al., 2000).

### Gastrointestinal Problems

Gastrointestinal (GI) problems, including constipation, abdominal pain, bloating, diarrhea, and nausea, are estimated to impact between 9 and 70% of children with ASD (Buie et al., 2010). The relationship between GI

impairments and ASD is unclear, but recent work suggests a shared genetic etiology for atypical neuronal development in both the central nervous system, impacting brain development and the enteric nervous system, impacting gut motility, that might lead to comorbid challenge in a subset of children with ASD (Bernier et al., 2014). Another avenue of investigation suggests anxiety may be playing a role. Studies indicate significantly higher rates of anxiety in children with GI problems (Mazurek et al., 2013; Nikolov et al., 2009).

### Sleep Problems

Sleep problems are commonly reported in children with ASD with reports indicating that between 44 and 83% of children with ASD have severe sleep problems (Allik, Larsson, & Smedje, 2006a, 2006b; Goldman, Richdale, Clemons, & Malow, 2012; Goldman et al., 2009; Krakowiak, Goodlin-Jones, Hertz-Picciotto, Croen, & Hansen, 2008; Owens, Spirito, McGuinn, & Nobile, 2000; Patzold, Richdale, & Tonge, 1998; Richdale, 1999; Richdale & Prior, 1995). Reported sleep problems include difficulty falling and staying asleep, early-morning waking, and overall shortened sleep times. These problems are most often reported in children under 5 years of age, but sleep problems persist for many despite age related improvements. The cause of sleep disturbances in ASD remains unknown, but disruptions to melatonin synthesis (Richdale, 1999), alterations in melatonin regulation (Rapin & Katzman, 1998), disruptions to arousal (Richdale & Prior, 1995), and anxiety (Wagner, 1991) have been proposed as possible etiologies. Regardless of the origin, actigraphy, an objective measure of sleep, indicates that children with ASD sleep shorter amounts than children with typical development or developmental delays (Goodlin-Jones, Tang, Liu, & Anders, 2008). Treatment for sleep disturbances in ASD, including supplemental melatonin, has been shown to be effective through both actigraphy and parent report of sleep and behavior (Doyen et al., 2011; Malow, Adkins, et al., 2012; Malow, Byars, et al., 2012; Rossignol & Frye, 2011). Further, treatment for sleep difficulties appears to also improve related symptoms such social interaction, repetitive behaviors, affective problems, oppositional and aggressive behavior, and inattention/hyperactivity symptoms (Goldman et al., 2009, 2012; Malow, McGrew, Harvey, Henderson, & Stone, 2006). Given the frequent concerns regarding sleep problems in ASD, the effectiveness of treatment, and the associated improvements in other behaviors

following sleep treatment, it is important that assessment and management of sleep concerns be a routine part of clinical care for children with ASD.

### Regression

While development can progress atypically from very early in life in ASD, symptoms may also appear following a loss of skills previously developed. This regression is estimated to occur between 20% and 47% of children with ASD (Goldberg et al., 2003; Lord, 1995; Rapin & Katzman, 1998; Rutter & Lord, 1987) with typical regression onset between 18 and 24 months of age (Davidovitch, Glick, Holtzman, Tirosh, & Safir, 2000; Fombonne & Chakrabarti, 2001; Kurita, 1985, 1996; Rapin & Katzman, 1998; Rutter & Lord, 1987; Shinnar et al., 2001; Wilson, Djukic, Shinnar, Dharmani, & Rapin, 2003). Some reports suggest that individuals with regression have poorer outcomes (Luyster et al., 2005) while others have found that there are no differences between children with and without regression on language, cognitive, or behavioral outcomes at age 3 to 4 years (Werner & Dawson, 2005). Loss of language skills is most often reported but can include social engagement, cognitive ability, and adaptive skill loss. Any loss of skills warrants immediate evaluation (Filipek et al., 1999).

### EARLY IDENTIFICATION

Most parents report concerns with their child with ASD's development by 18 months of age (Howlin & Asgharian, 1999; S. Rogers, 2001), but the mean age of diagnosis across the United States ranges from 38 to 120 months (Daniels & Mandell, 2013) with an average age of 3.1 years (Mandell, Novak, & Zubrisky, 2005). However, the age at which a child receives a diagnosis is decreasing as a result of increased awareness, improved screening techniques, and more expertise; earlier identification is important for intervening earlier and therefore improving outcomes.

Established practice parameters have helped to improved screening of ASD in young children. In 2000, the American Academy of Neurology issued practice parameters outlining a two-tiered screening approach for ASD. The first level focuses on the identification of children at risk for atypical development (followed by identifying those specifically at risk for ASD) through routine developmental surveillance conducted at all well-child visits. The second level entails formal diagnostic procedures conducted by expert evaluators focusing on the diagnosis

**TABLE 3.2 Red flags indicating need for further evaluation for ASD**


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|                                                                                              |
|----------------------------------------------------------------------------------------------|
| Further developmental evaluation is warranted if a child fails to meet any of the following: |
| Babbling by 12 months                                                                        |
| Gesturing (e.g., pointing, waving bye) by 12 months                                          |
| Using single words by 16 months                                                              |
| Using spontaneous (not echoed) two-word phrases by 24 months                                 |
| OR if there is a loss of language or social skills at any age                                |

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of ASD (Filipek et al., 2000). Importantly, these parameters stipulated that should a child fail to meet certain milestones or demonstrate a loss of language or social skills at any time, then further evaluation is warranted (Table 3.2). The American Academy of Pediatrics clarified that in addition to routine level 1 surveillance to identify children at risk for developmental delay, specific screening using an autism specific screening tool should be utilized at 18 and 24 months of age regardless of any identified risks (Johnson, Myers, & American Academy of Pediatrics Council on Children With Disabilities, 2007).

Several screening measures for infants at level 1 screening have been developed that meet the American Academy of Pediatrics' recommendations. Some of these include the modified CHAT (M-CHAT), the Pervasive Developmental Disorders Screening Test, and the First Year Inventory and Infant Toddler Checklist. The M-CHAT (Robins et al., 2001) and Pervasive Developmental Disorders Screening Test (Siegel, 2004) are both level 1 questionnaire-based screening instruments designed to quickly capture parent report of their toddlers. The First Year Inventory (Baranek, Watson, Crais, & Reznick, 2003) lowers the age boundary to 12 months for parent report in level 1 screening. The Screening Tool for Autism in Toddlers (Stone, Coonrod, & Ousley, 2000) and the Communication and Symbolic Behavior Scales Developmental Profile (CSBS; Wetherby & Prizant, 2002) are both interactive tools for use in level 2 screening. Through 20-minute, play-based interactions, clinicians can gather information regarding the presence of ASD and generate targets for intervention. The CSBS also includes a general developmental screener, the Infant Toddler Checklist, and a follow-up caregiver questionnaire in addition to the play based interaction.

There is a considerable increase in the ability to diagnose autism at younger ages and research to indicate a diagnosis can be reliably diagnosed by 20 to 24 months of age (Baron-Cohen et al., 1996; Rogers, 2001). Further, studies suggest that early diagnoses remain stable over time such that diagnoses made in the first two years of life are retained by age 3 (Baron-Cohen et al., 1996; Lord, 1995; Stone et al., 1999). These findings underscore the importance of early

detection, followed by diagnosis so that intervention can begin as early in development as possible.

## EPIDEMIOLOGY

### Prevalence

Once thought to be a rare disorder, ASD is now one of the most common developmental disorders. In the 1980s, the prevalence of ASD was estimated to be 4 per 10,000 individuals (Ritvo et al., 1989), but estimates currently suggest more than 1% of children in the United States (1 in 68) are affected (Centers for Disease Control and Prevention, 2014). Multiple studies have reported an increase in prevalence of ASD diagnosis (Cavagnaro, 2009; Centers for Disease Control and Prevention, 2009, 2012; Hertz-Picciotto & Delwiche, 2009; King & Bearman, 2009; Newschaffer, Falb, & Gurney, 2005b) although it remains unclear if the increase in diagnosis is strictly related to improved identification and broadening definitions or if there is a true increase in prevalence (Rice et al., 2012). A meta-analysis spanning 20 studies between 1966 and 1997 revealed that the prevalence rate for children born before 1970 was 4.7 cases per 10,000 while the rate for those born in 1970 or later was 11.2 per 10,000 (Gillberg & Wing, 1999). Similar to the recent CDC studies, this analysis yielded an estimated annual increase in autism diagnoses of 3.8 percent per year, mirroring rising international prevalence rates (Fombonne, 2009). Current research examining the multiple ASD risk factors, including genetic and prenatal and early postnatal environmental factors and their interaction, that contribute to risk for autism may illuminate the cause of the changing diagnostic rates (Newschaffer et al., 2007).

### Gender

Males are diagnosed at approximately a 4:1 rate relative to females (Fombonne, 2009). A variety of mechanisms have been proposed, including endogenous hormones (Knickmeyer & Baron-Cohen, 2006; Knickmeyer et al., 2006), intrauterine exposures that impact male but not female growth (Hertz-Picciotto et al., 2006), and X-chromosome inactivation (Nagarajan et al., 2008). More recently it has been demonstrated that females have a higher mutational burden, more deleterious structural mutations in autosomal chromosomes, than males, suggesting increased numbers of genetic events, not linked to variants on the X chromosome, contribute to the development of ASD in females (Jacquemont et al., 2014).

Finally, it has been suggested that the diagnostic tools may be biased toward identifying ASD in males (Rivet & Matson, 2011), which underscores the limited research on the diagnosis of females with ASD (Rutter, Caspi, & Moffitt, 2003).

### Sociocultural Influences

Socioeconomic status (SES) and race have both been associated with ASD diagnosis. In European countries supporting population wide surveillance and universal health care, SES does not appear to be a factor in the prevalence of ASD; however studies conducted in the United States indicate an underascertainment of families with low SES backgrounds (Baird et al., 2006; Fombonne, Du Mazaubrun, Cans, & Grandjean, 1997; Larsson et al., 2005). Race is also associated with ASD. Differential prevalence rates of ASD have been observed based on ethnicity with the odds of having an ASD being 57% lower for non-Hispanic, African American children, than non-Hispanic, Caucasian children (Kogan et al., 2009). The age at which a child is diagnosed also varies with ethnicity. One study suggested that on average, Caucasian children are diagnosed 1.5 years earlier than African American children, and 2½ years earlier than Latino children (Mandell, Listerud, Levy, & Pinto-Martin, 2002). The association between SES and ASD is likely due to reduced availability of diagnostic and treatment resources for families with lower SES (Kogan et al., 2009) as opposed to an association with etiology. Similarly, it is likely other factors driving the association between race and ASD. For example, it has been suggested that parents of certain ethnic backgrounds may emphasize disruptive or other behaviors during assessment rather than social oddities, leading to a reduced likelihood of an ASD diagnosis (Mandell et al., 2002). Alternatively, it has been proposed that general prejudices held by the clinician can lead to misguided expectations about the probability of ASD occurring in different ethnicities (Balsa, McGuire, & Meredith, 2005). Ultimately, the existing literature suggests that ASD does not discriminate based on racial or socioeconomic background.

### ETIOLOGY

No one clear cause of ASD has been identified, although significant advances have been made toward genetic susceptibility factors and potential environmental triggers. It is generally considered that ASD has a variety of contributors including genetic and environmental factors.

### Genetic Risk Factors

#### Twin Studies

The first clear indication that ASD had a strong genetic component was from Folstein and Rutter's (1977) study of twins with ASD. In their same-sex epidemiological study they found a 36% concordance rate for MZ twins and a 0% rate for DZ twins. Subsequent twin studies have replicated these findings with concordance rates ranging from 60–96% in MZ twins compared to 0–23% in DZ twins (Bailey et al., 1995; Ritvo, Freeman, Mason-Brothers, Mo, & Ritvo, 1985; Steffenburg et al., 1989). However, a recent California based population study of twins with ASD observed that common environmental factors may contribute approximately 55% of the ASD risk with genetic heritability accounting for less than 40 percent (Hallmayer et al., 2011). This recent study highlights that while genetic factors play an important role, the role may be lower than previously estimated.

#### Family Studies

The study of family members of an individual with ASD has illuminated the potential role common genetic variation in the susceptibility to ASD. The broader autism phenotype has been observed in family members of individuals with ASD. Defined as having similar traits to ASD (such as aloofness, rigidity of thought, repetitive interests), but to lesser degree, the broader autism phenotype was identified in a large-scale study in 7.5% of relatives of individuals with ASD while in only 2.5% of relatives with Down syndrome (Pickles et al., 2000). Similarly, the broader autism phenotype has been observed in 12–20% of siblings of children with ASD but in only 2% of siblings of children with developmental delays and observed more often in parents of children with ASD relative to parents of typical children or children with delays (Bailey et al., 1995; Baron-Cohen & Hammer, 1997; Bernier, Gerdts, Munson, Dawson, & Estes, 2012; Dawson, Webb, Wijsman, et al., 2005; Gerdts, Bernier, Dawson, & Estes, 2013; Losh et al., 2009). The observation of a broader phenotype suggests that relatives of individuals with ASD may be demonstrating a lesser variant a result of shared genes (Baron-Cohen & Hammer, 1997).

A large study of infant siblings of children with ASD indicated that 18.7% of younger siblings of individuals with ASD went on to develop ASD with risk increasing the more older siblings with ASD in the family (Ozonoff et al., 2011). In contrast, only approximately .18% of second-degree relatives and .12% of third-degree relatives of an individual with ASD also have ASD (DeLong & Dwyer, 1988;

Jorde et al., 1991; Pickles et al., 1995; Szatmari, Volkmar, & Walter, 1995). The differential rates of ASD in extended family members relative to siblings suggest that many genes may contribute to ASD (Pickles et al., 1995).

### Genetic Findings

Early molecular studies of ASD focused on linkage analysis to identify potential genomic regions harboring ASD susceptibility gene or genes. Through these studies many chromosomal regions were identified, yet few were replicated reflecting the heterogeneity in genetic contributions (e.g., Barrett et al., 1999; IMGSAC, 1998; Philippe et al., 1999; Risch et al., 1999) and no single gene region has been reliably identified (Abrahams & Geschwind, 2008).

Research suggests that there is no single gene that accounts for ASD, but rather that there are multiple genes and structural rearrangements, each one a risk factor that may alone or in concert with other genetic variation, contribute to ASD (Folstein & Rosen-Sheidley, 2001). Indeed, approximately 10–30% of ASD cases have an identified genetic cause through an observable chromosomal structural rearrangement, called copy number variation (CNV), or gene mutation (Abrahams & Geschwind, 2008; Krumm, O’Roak, Shendure, & Eichler, 2014). Among individuals with an identified genetic contributor, the most common contributors are CNVs in regions 15q11–13 and 16p11.2, CNVs at each region accounting for only approximately 1% of cases (Miles, 2011). However, complicating the picture is that individuals with CNVs in these regions, such as 16p11.2, ascertained for the presence of the event, do not always develop ASD. In fact, ASD is only observed in 23% of individuals with 16p11.2 deletion (Hanson et al., 2014). With the advent of exome and targeted sequencing the discovery of disruptive gene mutations has advanced the understanding of genetic risk factors. Indeed, exome sequencing has yielded findings that close to 1,000 different genes may contribute to ADS (O’Roak, Vives, Girirajan, et al., 2012; Sanders et al., 2012). Among the most prominent are CHD8, DYRK1A, GRIN2B, PTEN, TBR1, TBLXR1, with mutations to these 6 genes likely contributing to 1% of all ASD cases (O’Roak, Vives, Fu, et al., 2012). Indeed, CHD8 appears to be the most common single gene mutation contributing to ASD and represents a distinct subtype of ASD with characteristic facial features, macrocephaly, and gastrointestinal disturbances (Bernier et al., 2014). Many other genes have been linked to ASD including ADNP (Helsmoortel et al., 2014), SCN2A (State & Sestan, 2012), SHANK2, SHANK3, and NLGN4 (Abrahams & Geschwind, 2008), which highlights the varied impacts on the development of the brain given

the wide role on synaptic adhesion, chromatin modification, ion channel development, and the neurotransmitter receptors in which these genes are involved.

Our understanding of the genetics of ASD has advanced considerably since early twin studies, and these advances indicate ASD does not follow a simple pattern of inheritance and instead is the result of multiple genetic variants and mutations impacting a multitude of genes. Given the increased availability of high throughput sequencing and reduced cost for sequencing, a shift in the approach to studying the genetics of ASD has been proposed which entails the leveraging of technology to genetically classify individuals with ASD and then recontact and follow up for comprehensive evaluation to identify relevant behavioral subtypes and targets for treatment and intervention (Stessman, Bernier, & Eichler, 2014).

### Environmental Risk Factors

#### Prenatal Risk Factors

A wide range of pre and perinatal complications has been associated with increased risk for ASD. Although no single prenatal factor has been associated with the etiology of ASD (Gardener, Spiegelman, & Buka, 2009), a large, population based study of environmental contributions of ASD suggested that fever during pregnancy increased the risk for ASD (Zerbo, Iosif, Delwiche, Walker, & Hertz-Picciotto, 2011). The proposed mechanism for this increased risk following maternal infections is that the immune response and maturational processes at critical developmental periods are disrupted (Hertz-Picciotto et al., 2008). It has also been reported that mothers with certain medical conditions, including diabetes (type 2 or gestational), chronic hypertension, or pre-pregnancy obesity was associated with increased risk for autism (Krakowiak et al., 2012). The authors hypothesize that metabolic disruption results in increased ASD risk. Rodent and non-human primate models mirror these findings of increased ASD susceptibility following maternal immune activation (M. D. Bauman et al., 2013; Malkova, Yu, Hsiao, Moore, & Patterson, 2012).

Other prenatal factors associated with ASD risk include exposures in utero to a variety of agents including thalidomide (Rodier, 2002; Stromland, Nordin, Miller, Akerstrom, & Gillberg, 1994), misoprostol (Bandim, Ventura, Miller, Almeida, & Costa, 2003), and valproic acid (Moore et al., 2000; Rasalam et al., 2005; Stromland et al., 1994). Across all of these exposures, the mechanism is based on reported observations of the teratogenic effects of these substances on neurodevelopment (Landrigan, 2010). More

recently, air pollution (Volk, Hertz-Picciotto, Delwiche, Lurmann, & McConnell, 2011; Volk, Lurmann, Penfold, Hertz-Picciotto, & McConnell, 2013) and agricultural pesticides (Eskenazi et al., 2007; Roberts et al., 2007; Shelton et al., 2014) have been associated with increased risk. The hypothesized mechanisms for these environmental pollutants are that thyroid hormones, which have been implicated in neurodevelopmental problems (Porterfield, 1994), are disrupted (Roberts et al., 2007) and that sex steroids are impacted, which have been proposed to account for the observed male to female ratio bias in ASD (Baron-Cohen, Knickmeyer, & Belmonte, 2005; Knickmeyer et al., 2006).

Use of prenatal folic acid supplements around the time of conception has been associated with decreased risk for autism (Suren et al., 2013). Research also suggests that the MTHFR C677T polymorphism is associated with increased ASD risk, and periconceptional folic acid may reduce ASD risk in those with MTHFR 677C>T polymorphism (Pu, Shen, & Wu, 2013).

### *Perinatal Risk Factors*

A broad range of perinatal complications have been proposed to increase ASD risk, including breech presentation, umbilical cord complications, fetal distress, birth injury or trauma, multiple birth, maternal hemorrhage, summer birth, low birth weight, small for gestational age, congenital malformation, low Apgar scores, feeding difficulties, meconium aspiration, neonatal anemia, ABO or Rh incompatibility, and hyperbilirubinemia (Gardener, Spiegelman, & Buka, 2011). The relationship between the complication and ASD may be a function of the hypoxia associated with the event, which can impact central nervous system and later development. Elevated risk has also been found with non-elective C-sections, but it is likely a pre-existing condition necessitating the surgery as opposed to the surgery itself that increases ASD risk (Krakowiak et al., 2012). Additionally, a number of perinatal complications have not been found to associate with ASD, including use of anesthesia, post-term birth, and physical characteristics such as high birth weight and head circumference. Increased risk for autism has been associated with induced or augmented childbirth (Gregory, Anthopolos, Osgood, Grotegut, & Miranda, 2013). Further research is needed to explain this association which could be related to underlying pregnancy conditions required need to induce/augment as opposed an independent contributor of ASD risk.

In general, challenges to clarifying the extent and effects of these perinatal complications concern the interaction between birth complications and prenatal problems, making it difficult to determine the independent contribution of any of these factors to conferring ASD risk. For example, C-sections are more common in pregnancies with abnormal fetal presentation, fetal distress, and multiple births, preventing a clear understanding of the independent effect of any birth complication. Further research in this arena is needed.

### **Gene–Environment Interactions**

An important area of investigation exists at the intersection between genetics and the environment, as it is likely that the etiology of many individuals with ASD is a function of both genetic and environmental factors (Newschaffer et al., 2007). This perspective is supported by the observation of abnormal neuroinflammation following infection in individuals with ASD and the potential role of gene disruptions related to DNA repair and immune and inflammatory response (Herbert et al., 2006) as well as the finding that individuals with a promoter variant to the MET gene who are prenatally exposed to environmental toxins are at increased risk above prenatal exposure and having the genetic variant alone (Volk et al., 2014). Another gene environment interaction is reflected in the finding that increased paternal age confers ASD risk potentially by the alteration in the germ cells as a result of accumulated exposure to environmental toxins over the lifetime (Hultman, Sandin, Levine, Lichtenstein, & Reichenberg, 2011). Given the recent advances in genetics of ASD and the scrutiny of the environment contributions to the disorder, the examination of the intersection of the two factors will be at the forefront of the research agenda in the coming years.

### **BRAIN STRUCTURE AND FUNCTION**

Studies employing brain autopsies, structural and functioning imaging, and electroencephalography all converge to indicate atypical brain structure and function in ASD. Neuroanatomical findings, in general, conclude that ASD is marked by structural enlargement in some brain regions and reduction in other areas (see Koenig, Tsatsanis, & Volkmar, 2001, for a review). Although seemingly contradictory, these findings suggest that across the many etiologies of ASD, abnormal cell growth during the early

stages of prenatal and postnatal brain development may be the common underlying etiological mechanism. That is, in contrast to typical brain development in which neurons proliferate and become interconnected and then are pruned through experience and development, in ASD, the process of neural growth and pruning may be disrupted resulting in size differences across neuroanatomical structures (Minshew, 1996). Imaging studies show increased overall brain volume (Courchesne et al., 2001; Hazlett et al., 2011; Piven, Arndt, Bailey, & Andreasen, 1996; Piven et al., 1995; Schumann et al., 2010; Sparks et al., 2002) and autopsy studies report increased brain weight (Bailey et al., 1998; Bauman & Kemper, 1994). These findings are supported by studies of head circumference that demonstrates rapid, early growth in ASD (Chawarska et al., 2011; Courchesne, Campbell, & Solso, 2011; Courchesne & Pierce, 2005a; Dawson et al., 2007; Elder, Dawson, Toth, Fein, & Munson, 2008). Enlargement of both white and gray matter appears to account for this rapid growth (Hazlett et al., 2011; Schumann et al., 2010), but a large study of brain growth in teens and young adults suggests that atypical decline may occur at these ages as well (Courchesne et al., 2011). In addition to the observation of atypical brain growth in ASD, several specific brain regions have been implicated, including the cerebellum, the limbic system,

several systems in the cerebral cortex, and corpus callosum. The specific regions and structures are outlined in Table 3.3.

### Cerebellum

Studies employing different experimental methods have identified atypical structure and function of the cerebellum in ASD. Structural MRI studies report enlarged cerebellar volume (Carper & Courchesne, 2000; Courchesne, Redcay, Morgan, & Kennedy, 2005), magnetic resonance spectroscopy (MRS) studies indicate decreased concentrations of neural substrates indicating diminished cerebellar functioning (Chugani, Sundram, Behen, Lee, & Moore, 1999; DeVito et al., 2007), and functional MRI studies demonstrate atypical activity in the cerebellum during attention and motor tasks (Allen, Müller, & Courchesne, 2004). Similarly, postmortem studies in ASD have found a loss of Purkinje and granule cells in the region of the cerebellum that receives auditory and visual information (Bailey et al., 1998; Bauman & Kemper, 1985; ML Bauman & Kemper, 1988; Carper & Courchesne, 2000; Palmen, van Engeland, Hof, & Schmitz, 2004; Whitney, Kemper, Bauman, Rosene, & Blatt, 2008). Two proteins implicated in cell migration and pruning, Reelin and Bcl-2, are reportedly decreased in cerebellar tissue (Fatemi, Stary, Halt, & Realmuto, 2001).

**TABLE 3.3 Brain Regions and Structures Implicated in in ASD and Their Purported Function**

| Brain area                     | ASD-related domain                                       | Proposed function                                                 | Citation                                                |
|--------------------------------|----------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------|
| Ventromedial prefrontal cortex | Social communication                                     | Social decision making                                            | Carper and Courchesne (2005)                            |
| Orbitofrontal cortex           | Social communication                                     | Adaptive learning through stimulus–reinforcement association      | Girgis et al. (2007)                                    |
| Anterior cingulate cortex      | Social communication                                     | Social orienting, initiation, motivation, goal-directed behaviors | Mundy (2003)                                            |
| Inferior frontal gyrus         | Social communication                                     | Imitation, language production                                    | Dapretto et al. (2006)                                  |
| Posterior parietal cortex      | Social communication                                     | Imitation, understanding goals and intentions                     | Williams et al. (2006)                                  |
| Fusiform gyrus                 | Social communication                                     | Face perception and processing                                    | Schultz et al. (2000)                                   |
| Superior temporal sulcus       | Social communication                                     | Biological motion perception, understanding goals and intentions  | Pelphrey and Carter (2008)                              |
| Amygdala                       | Social communication                                     | Assignment of emotional valence, emotional learning               | Baron-Cohen et al. (2000)                               |
| Basal ganglia                  | Repetitive and restricted behaviors                      | Compulsive behaviors, repetitive motor behaviors, routines        | Langen, Durston, Staal, Palmen, and van Engeland (2007) |
| Substantia nigra               | Social communication/Repetitive & restricted behaviors   | Reward, reinforcement, and motivation                             | Lewis, Tanimura, Lee, and Bodfish (2007)                |
| Thalamus                       | Social communication/Repetitive and restricted behaviors | Sensory-motor integration                                         | Tsatsanis et al. (2003)                                 |
| Cerebellum                     | Social communication/Repetitive and restricted behaviors | Engagement and disengagement of attention                         | Courchesne (1997)                                       |
| Hippocampus                    | Social communication                                     | Learning and memory                                               | Bauman and Kemper (2005)                                |

The findings of atypical cerebellar functioning mirror some of the atypical behaviors and skills observed in ASD that have been associated with the cerebellum. For example, many individuals with ASD have difficulties with sequencing language, motor movements, and shifting attention. These abilities are partially mediated by the cerebellum. Importantly, the ability to shift attention is critical for joint attention, theory of mind, and the ability to transition from one activity to another, all symptoms associated with ASD (Carper & Courchesne, 2000; DiCicco-Bloom et al., 2006; Gillberg, 1999).

### The Amygdala

Evidence for disruptions in the amygdala and hippocampus in ASD follows from early findings of decreased neuronal cell size in these structures (Bauman & Kemper, 1988, 2005; Schumann & Amaral, 2006). Studies have found increased amygdala size (relative to total brain volume) in ASD with more severe symptoms (Schumann, Barnes, Lord, & Courchesne, 2009; Schumann et al., 2004; Sparks et al., 2002) although findings reveal a relationship with age and the amygdala (Stanfield et al., 2008). Size differences are not found in older children and adults (Aylward et al., 1999; Haznedar et al., 2000; Nacewicz et al., 2006) suggesting the growth observed in middle childhood in typical development does not occur in ASD (Amaral, Schumann, & Nordahl, 2008). Interestingly, enlargement of the amygdala in toddlerhood is predictive of more significant impairment trajectory in ASD between 3 and 6 years of age (Munson et al., 2006). Two competing hypothetical mechanisms have been proposed to account for the amygdala findings in ASD: either fewer neurons are generated in early development or an excessive number are but there is subsequent elimination of those neurons (Amaral et al., 2008).

### Corpus Callosum

The wide band of nerve fibers linking the left and right hemispheres of the brain appears to be reduced in size in children and adults with ASD (Alexander et al., 2007; Boger-Megiddo et al., 2006; Egaas, Courchesne, & Saitoh, 1995; Hardan, Minshew, & Keshavan, 2000; Manes et al., 1999; Piven, Bailey, Ranson, & Arndt, 1997; Vidal et al., 2006). This observed size difference in these white matter tracts in ASD supports the theory that disruptions to cortical connectivity may play a role in ASD. Increased white matter has been observed throughout the brain with the frontal lobe white matter showing the greatest increase

(Courchesne et al., 2001; Hazlett et al., 2005; Herbert et al., 2004) implicating a deficit in regional or hemispheric brain communication (Belmonte et al., 2004; Courchesne & Pierce, 2005b; Just, Cherkassky, Keller, & Minshew, 2000; Penn, 2006; Rippon, Brock, Brown, & Boucher, 2007).

### Cortical Regions

Imaging and neurophysiological studies indicate abnormal development and functioning of a number of cortical regions. Consistent atypicality has been reported in the prefrontal cortex (Carper & Courchesne, 2000; Courchesne et al., 2005; DeVito et al., 2007). Dysfunction and thinning in the frontal lobes has been associated with theory of mind (F. Happe et al., 1996), imitation impairments (Dapretto et al., 2006), and executive functioning (Gilbert, Birch, Brindley, Frith, & Burgess, 2008). Other cortical regions have been proposed to be involved in ASD including the fusiform gyrus, implicated in functional imaging and neurophysiological studies of face processing (Dawson et al., 2002; Dawson, Webb, et al., 2004; Kleinhan, Muller, Cohen, & Courchesne, 2008; Schultz, 2005; Webb et al., 2006), and the superior temporal sulcus, implicated in perception of biological motion and intention understanding (Pelphrey, Morris, & McCarthy, 2005; Pelphrey, Adolphs, & Morris, 2004). The parietal lobe has been implicated as well through imaging and neurophysiological studies consistent with the proposal of mirror neuron dysfunction in ASD (Bernier et al., 2007; Oberman & Ramachandran, 2007; Williams et al., 2006).

## INTERVENTIONS AND TREATMENT

Pioneering work in the '60s and '70s adopted behavioral principles in the treatment of ASD (Berkowitz & Graziano, 1972; Hingtgen, Coulter, & Churchill, 1967; Leff, 1968; Lovaas, Schreibman, & Koegel, 1974; Mazuryk, Barker, & Harasym, 1978) and following the publication of Lovaas' (1987) controlled intervention trial based on the principles of applied behavioral analysis (ABA), the notion that ASD could be treated was ushered into the professional and parent community. In the years since this early work, major advances in ASD intervention have been made by building upon behavioral principles, incorporating naturalistic approaches, and embracing a developmental framework. Additionally, myriad psychopharmacological as well as complementary and alternative approaches have been developed and are currently in use in the treatment

of autism interventions are still lacking (Warren et al., 2011). Despite this, with the advent of intervention there have been significant improvements in the outcomes for children with ASD with a minority of children actually losing symptoms altogether and achieving functioning within the normal range (Fein et al., 2013).

### Behaviorally Based Intervention Models

Interventions based on ABA principles have garnered the most empirical support in the treatment and management of ASD (National Research Council, 2001). Multiple studies have replicated Lovaas's (1987) findings of efficacy (Cohen, Amerine-Dickens, & Smith, 2006; Howard, Sparkman, Cohen, Green, & Stanislaw, 2005; Sallows & Graupner, 2005) and long-standing gains have been observed (McEachin, Smith, & Lovaas, 1993). Indeed, one review of outcomes of 373 children in 13 studies indicated significant gains in cognitive ability following ABA based treatment (Reichow & Wolery, 2009). ABA based treatments have been shown to improve abilities across ASD domains including social communication and challenging behavior (Cohen et al., 2006; Horner, Carr, Strain, Todd, & Reed, 2002; McConnell, 2002; Sallows & Graupner, 2005).

Improved understanding of child development and the importance of generalizability of skills have led to the establishment of several naturalistic developmental models. Pivotal Response Training (PRT) is a naturalistic model that focuses on "pivotal" behaviors, such as motivation. Studies of PRT indicate improvement in social communication and play following intervention (Boettcher, 2004; Koegel, Carter, & Koegel, 2003; Koegel, Koegel, Shoshan, & McNeerney, 1999; Koegel, Camarata, Koegel, Ben-Tall, & Smith, 1998; Schreibman, Stahmer, & Pierce, 1996). The Social Communication, Emotional Regulation, and Transactional Support model (SCERTS model) focuses on the specific learning styles to address the core deficits of children with ASD (Prizant, Wetherby, Rubin, & Laurent, 2003; Prizant, Wetherby, Rubin, Laurent, & Rydell, 2005). Kasari and colleagues (2006) have developed joint attention training to improve language and overall outcomes in ASD by addressing this precursor skill to so many core deficits in ASD. A randomized controlled trial of the 6-week intervention suggests improvements following treatment that persist 5 years later (Kasari, Gulsrud, Freeman, Paparella, & Hellemann, 2012) and that can be observed when administered in community preschool programs (Lawton & Kasari, 2012).

The Early Start Denver Model (ESDM), a downward extension of the Denver Model (Rogers et al., 2006; Rogers & Lewis, 1989), is a comprehensive developmental model based on ABA principles. A randomized, controlled trial of 48 toddlers demonstrated significantly greater improvement across cognitive, language, social, and adaptive domains for the ESDM group relative to the children receiving treatments in the community (Dawson et al., 2010). Further, following the 2 years of intensive intervention the children receiving the ESDM intervention showed normalization of brain activity in response to viewing faces and houses using EEG (Dawson et al., 2012), suggesting that this model significantly impacted not only behavior but also neurological functioning.

### Pharmacological Intervention

Between 30 and 70% of children with ASD are prescribed medication in the course of their treatment (Aman, Lam, & Collier-Crespin, 2003; Aman, Lam, & Van Bourgondien, 2005; Esbensen, Greenberg, Seltzer, & Aman, 2009) although there are currently no medications that have demonstrated efficacy for addressing the core symptoms in ASD. The two FDA approved medications showing efficacy in ASD, risperidone and aripiprazole, target challenging behaviors and hyperactivity but not ASD itself (Marcus et al., 2009; Owen et al., 2009; Scahill et al., 2012). A comprehensive review of medical treatments in ASD indicates that most pharmacological interventions have limited evidence for their efficacy and those that do have been supported by private pharmaceutical companies. Further, no studies compare medical interventions to behavioral interventions or include combinations of both, even though most children with ASD receive both types of treatment (McPheeters et al., 2011).

Despite the limited extant literature, there are novel biomedical treatments under evaluation, such as arbaclofen, to address the core social deficits in ASD (Gurkan & Hagerman, 2012). Recent studies of oxytocin suggest promise (Domes et al., 2013) and indicate that even a single intranasal injection of oxytocin appears to normalize brain functioning and may therefore be an appropriate tool to bootstrap or enhance the effectiveness of behaviorally based interventions (Gordon et al., 2013).

### Complementary and Alternative Approaches

Complementary and alternative treatments (CAT), defined as "medical and health care systems, practices, and products that are not generally considered to be part of

conventional medicine" (National Center for Complementary and Alternative Medicine, 2012), are commonly used in ASD with reports of between 32 and 80% of families reporting using or having used CAT to address ASD symptoms (Harrington, Rosen, Garnecho, & Patrick, 2006; Levy & Hyman, 2003; Levy, Mandell, Merhar, Ittenbach, & Pinto-Martin, 2003; Wong & Smith, 2006). This diverse group of treatments includes biomedical approaches, such as vitamins, supplements, diets, and procedures such as neurofeedback; mind-body interventions, such as meditation; body-based methods, such as massage; energy based treatments, including reiki; or alternative care systems (e.g., homeopathy) and the primary reported rationale for applying these approaches relate to safety concerns or adverse medication side effects (Hanson et al., 2007). Despite the widespread availability and application of CATs, there is limited scientific evidence supporting the efficacy of these approaches due to a lack of controlled trials, rigorous protocols, adequate sample sizes, clear inclusion criteria, and double blind designs (Lofthouse, Hendren, Hurt, Arnold, & Butter, 2012).

Many CATs have few or no studies examining their efficacy providing insufficient evidence in which to evaluate them; however, some have been studied using more rigorous designs. Melatonin is among the best-studied CAT for ASD and despite limitations to study designs, sample sizes, and variability in approaches, the existing literature suggests the use of melatonin holds promise for addressing sleep problems in ASD (Rossignol & Frye, 2011). Another well-studied CAT is Secretin. It has consistently been shown to have no utility in the management of ASD (Krishnaswami, McPheeters, & Veenstra-Vanderweele, 2011). Following media reports of its efficacy in a small group of children with ASD and diarrhea, excitement over Secretin as a treatment preceded the research, and only after dozens of studies failing to show efficacy was Secretin removed from favor (Schreibman, 2005). Other treatments that have failed to demonstrate positive effects through randomized controlled trials include auditory integration therapy (Sinha, Silove, Wheeler, & Williams, 2004), facilitated communication (Mostert, 2001), hyperbaric oxygen therapy (Jepson et al., 2011), and gluten/casein-free diet (Millward, Ferriter, Calver, & Connell-Jones, 2008). Countless other approaches still have insufficient evidence with which to evaluate their effectiveness, including many specific vitamin supplements, immune therapies, and movement therapies (Lofthouse et al., 2012). In total, although few specific CATs have sufficient evidence for

effectiveness, it has been suggested that families work with their primary medical provider together to consider and perhaps utilize CATs that show potential promise, sensible rationale for effectiveness, fit with family's values, and show evidence for safety (Hendren, 2013).

## FUTURE DIRECTIONS

Major advances in our understanding of ASD have been made in the intervening years since Leo Kanner first defined the disorder in 1943. It is clear now that ASD can be conceptualized as a group of disorders with significant heterogeneity in terms of etiology, course, response to treatment, and outcome. Developmental research has illuminated the core social impairments in ASD and provided insight into the downstream effects of these deficits on clinical presentation. Studies have also demonstrated the viability and importance of detecting ASD very early in development and identified several genetic and environmental contributors and risk factors. Investigations in neuroanatomy and function indicate atypical growth and disruption to a range of structures and regions and the focus on intervention has shown that although all children with ASD benefit from intervention, some clear show rapid gains while others show slower response. Future directions in ASD research will expand on these gains. The identification of biomarkers to be used in early detection of ASD is a key focus of research and will allow for earlier application of intervention focused on the core deficits in ASD. By addressing these core deficits earlier, developmental trajectories can be normalized and outcomes optimized. Studies focused on etiology will address the heterogeneity of ASD by continuing to clarify the contributions of specific ASD associated genes and environmental risk factors. Future treatment research will likely entail more rigorous analysis of complementary and alternative approaches, the discovery of novel pharmacological agents or novel application of existing medications, careful identification of the mediators and moderators of treatment response for all treatment types, and the matching of treatments for specific, etiologically linked ASD subtypes. Ultimately the future holds the potential for bridging our understanding of early detection, etiology, and treatment such that biomarkers can be used to identify children at risk for ASD, so that relevant, targeted treatments based on etiological subgroups can be applied to improve outcomes for individuals with ASD.

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